Universality classes for purification in nonunitary quantum processes

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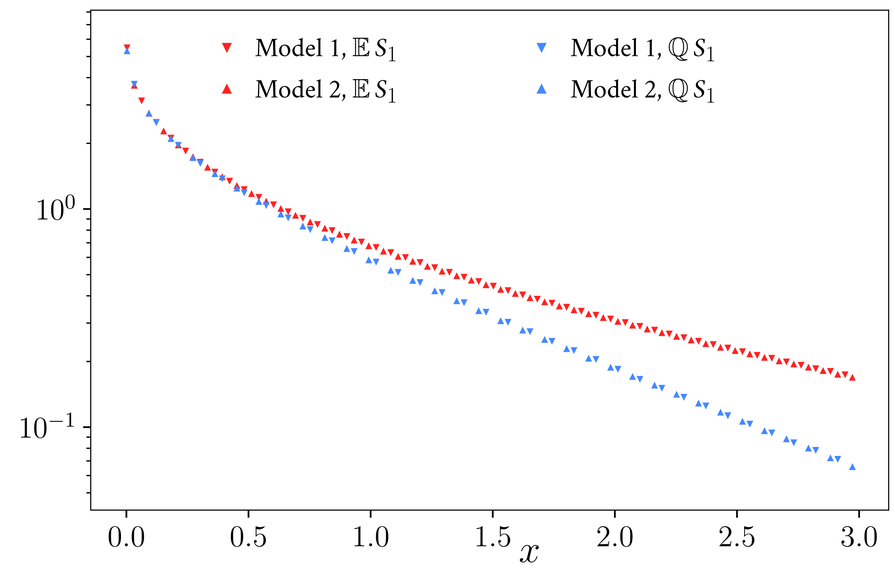
Abstract

We consider the universal aspects of two problems: (i) the singular value structure of a product $M_t = m_t m_{t-1} \dots m_1$ of many large independent random matrices, and (ii) the slow purification of a large number of qubits by repeated quantum measurements. Both processes are associated with the decay of natural measures of entropy as a function of time or of the number of matrices in the product. We argue that for a broad class of models, each process is described by universal scaling forms for purification and that (i) and (ii) represent distinct "universality classes" with distinct scaling functions. Using the replica trick, these universality classes correspond to effective one-dimensional statistical mechanics models for a gas of "kinks", representing domain walls between elements of the permutation group. This is an instructive low-dimensional limit of the effective statistical mechanics models for random circuits and tensor networks.



Model 1 and Goal

Model 1. A product of t random matrices acting on a q-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^q$

$$M_t := m_t m_{t-1} \dots m_1, \tag{1}$$

where each m_i independently comes from the *complex Ginibre ensemble*, namely

$$((m_i)_{\alpha\beta})_{1\le\alpha,\beta\le q,\ 1\le i\le t} \overset{i.i.d.}{\sim} \mathcal{N}(0,1). \tag{2}$$

Define the *unnormalized* and *normalized* density matrices $\check\rho$ and ρ by

$$\check\rho := M_t M_t^\dagger, \qquad \rho := \frac{\check\rho}{\operatorname{tr}\check\rho}. \tag{3}$$

Goal. Examine the large t, large q behavior of the expectation value $\mathbb{E}\, F$ of various functionals $F[\rho]$ of ρ (or $\check\rho$); In particular, the (information-theoretic) *Rényi* and *von Neumann entropies*

$$\mathbb{E}\, S_n := \mathbb{E}\left[\frac{1}{1-n}\ln\operatorname{tr}\rho^n\right], \qquad \mathbb{E}\, S_1 = \lim_{n\to1}\mathbb{E}\, S_n \equiv \mathbb{E}\left[-\operatorname{tr}\rho\ln\rho\right], \tag{4}$$

and the *quantum (Born rule) average* (see Model 2 for motivation in physics)

$$\mathbb{Q}\, S_n := \mathbb{E}\left[\frac{1}{1-n}\operatorname{tr}\check\rho\ \ln\operatorname{tr}\rho^n\right], \qquad \mathbb{Q}\, S_1 = \lim_{n\to1}\mathbb{Q}\, S_n \equiv \mathbb{E}\left[-\operatorname{tr}\check\rho\ \operatorname{tr}\rho\ln\rho\right]. \tag{5}$$

Model 1 – Main Results

1. Existence of a scaling regime. In the *scaling regime*, defined as $t, q \to \infty$ with a fixed ratio $t/q = x$, any function $\mathbb{E} f[\rho]$ depends only on $x = t/q$ (rather than depend on the full matrix products). For example, we have $\mathbb{E}\, S_n = F_n(x)$ for some function F_n.

2. Small x expansion. For small x, we have

$$\mathbb{E}\, S_1 = -\ln x + 1 - \gamma + \frac{11}{24}x^2 - \frac{1739}{2880}x^4 + \frac{329489}{181440}x^6 - \frac{83530439}{9676800}x^8 + O(x^{10}), \tag{6a}$$

$$\mathbb{Q}\, S_1 = -\ln x + 1 - \gamma + \frac{5}{24}x^2 - \frac{239}{2880}x^4 + \frac{3679}{36288}x^6 - \frac{2423279}{9676800}x^8 + O(x^{10}), \tag{6b}$$

where $\gamma \approx 0.577216$ is the Euler-Mascheroni constant.

3. Large x expression. For finite x, $\mathbb{Q}\, S_1$ follows from a determinantal point process

$$\mathbb{Q}\, S_1 = -\ln\left[2\sinh\frac{x}{2}\right] + 1 - \gamma + \int_0^\infty du \frac{\left(\lim_{R\to\infty} G_R(u,x)\right) - e^{-u}}{u^2}, \tag{7}$$

where

$$G_R(u,x) := \det\left(I_{j,k}(ue^{-x(R-1/2)})\right)_{j,k=0}^{R-1}, \tag{8a}$$

$$I_{j,k}(\tilde u) := e^{-x(j^2+k^2)/4}\sum_{h=0}^{\infty}\frac{e^{xh^2/2}(-\tilde u)^{h-(j+k)/2}}{(h-j)!(h-k)!} \tag{8b}$$

$$= \mathrm{i}^{|j-k|}e^{-\frac18 x(j-k)^2}\int_{-\infty}^{\infty}\frac{dw}{\sqrt{2\pi x}}e^{-\frac{w^2}{2x}}J_{|j-k|}\left(2e^{(w+(j+k)x/2)/2}\sqrt{\tilde u}\right). \tag{8c}$$

Model 2 and Further Results

Model 2. A *random quantum circuit* acting on a $q_{\text{tot}} = 2^{\mathcal{V}}$-dimensional Hilbert space $\mathcal{H} = \mathbb{C}^{q_{\text{tot}}}$, where each time step consists of (i) random unitary evolution followed by (ii) measuring $0 < r < 1$ fraction of the qubits in the computational basis.

This physical process defines a (Born rule) joint probability measure for the complete sequence of measurement outcomes up to time t (the *quantum trajectory*) in additional to the random matrix measure:

$$P(\text{outcomes}) = \operatorname{tr}\check\rho_t, \qquad \check\rho_t = (P_t U_t \cdots P_1 U_1)\rho_0(P_t U_t \cdots P_1 U_1)^\dagger. \tag{9}$$

Here P_i is a random orthogonal projector of rank q_s, and U_i a Haar random unitary matrix.

The Quantum average corresponding to this physical process is defined as

$$\mathbb{Q}\left[F[\rho_t]\right] := (q_{\text{tot}}/q_s)^t \times \mathbb{E}\left[\operatorname{tr}\check\rho_t F[\rho_t]\right]. \tag{10}$$

Further results. For *purification dynamics* ($\rho_0 = 1_{q_{\text{tot}}\times q_{\text{tot}}}$), the results 1–3 above holds, with

$$\frac{1}{q} := T_{1^N,1^{N-2}2} \simeq \frac{1}{q_s} - \frac{1}{q_{\text{tot}}}, \quad q_s := 2^{(1-r)\mathcal{V}}. \tag{11}$$

Method for Model 1 (Also applicable to Model 2)

Step 1: Exponential Generating Function (EGF). We turn to the calculation of

$$EGF_{N,k,n} := \mathbb{E}\left[\operatorname{tr}(\check\rho)^N e^{-k(n-1)S_n}\right] = \mathbb{E}\operatorname{tr}(\check\rho^n)^k\ \operatorname{tr}(\check\rho)^{N-nk}, \tag{12}$$

the entropies can be retrieved via the *replica trick*

$$\mathbb{E}\, S_n = \lim_{N\to0}\partial_k EGF_{N,k,n}\Big|_{k=0}, \qquad \mathbb{Q}\, S_n = \lim_{N\to1}\partial_k EGF_{N,k,n}\Big|_{k=0}. \tag{13}$$

Step 2: Transfer matrix expression. For Ginibre ensembles, the Gaussian average $\mathbb{E}[\cdot]$ can be straightforwardly carried out, and we get a transfer matrix expression

$$EGF_{N,k,n} = (T^{t+1})_{1^{N-nk}n^k,1^N}, \tag{14}$$

where T is an $N!\times N!$ matrix (up to an overall q^N factor)

$$T_{\sigma',\sigma} = q^{-|\sigma'\sigma^{-1}|}, \tag{15}$$

where $|\sigma'\sigma^{-1}|$ denotes the *Cayley distance* associated with the Cayley graph

$$G := (S_N, \text{Transpositions}). \tag{16}$$

Step 3: Seeing the scaling limit. At $q\to\infty$, we can do a large q expansion

$$T_{\sigma',\sigma} = \delta_{\sigma',\sigma} + A_{\sigma',\sigma}/q + o(1/q), \tag{17}$$

where A is the *adjacency matrix* of G. Then the scaling form emerges at large t:

$$T^t = \left(1_{N!\times N!} + A/q + o(1/q)\right)^t \xrightarrow{t\to\infty,\ q\to\infty,\ x:=t/q\text{ finite}} e^{xA}. \tag{18}$$

(For Model 2 we have instead $T_{1^N,\sigma} = \left(\frac{q_{\text{tot}}}{q_s}\right)^N\sum_{\tau\in S_N}\mathrm{Wg}(\tau^{-1}\sigma)(P^{\otimes 2N})_{\tau,1^N}$, with $(P^{\otimes 2N})_{\tau,1^N} = q_s^{N-|\tau|}$; here $\mathrm{Wg}(\sigma)$ is the Weingarten function, see Ref. [1].)

Step 4: Diagonalizing the adjacency matrix. We have the *Young diagram* expression

$$EGF_{N,k,n} \xrightarrow{\text{scaling limit}} \left(e^{xA}\right)_{1^{N-nk}n^k,1^N} = \frac{1}{N!}\sum_{\lambda\vdash N} d^\lambda\chi^\lambda(1^{N-nk}n^k)e^{x\nu^\lambda}, \tag{19}$$

where we defined the eigenvalues of the adjacency matrix A

$$\nu^\lambda := \frac{\chi^\lambda(1^{N-2}2)|\mathrm{Cl}(1^{N-2}2^1)|}{d^\lambda} = \frac12\left(\sum_j\lambda_j^2 - \sum_{j'}(\lambda_{j'}^\top)^2\right). \tag{20}$$

Step 5: Getting analytic expression. A polynomial conjecture leads to the small x expansion (6); a further resummation combined with the Andréief's equality leads to the large x expression (7)-(8). (Details in Sec. III E–G. of our paper.)

Results are universal

Universal physical picture – 1d kink gas: The quantity $\mathcal{Z} := EGF_{N,k,n}$ corresponds to a 1D classical spin model, for spins $\sigma_i \in S_N$, with fixed boundary conditions ($\sigma_{\text{final}} := 1^{N-nk}n^k$):

Existence of the scaling limit – a universal condition. We argue (Sec. II D) that the scaling limit exists whenever the fugacities of higher kinks μ is much smaller than that of the *elementary kink*:

$$T_{1^N,\mu} \ll q^{-1} := T_{1^N,(12)} \text{ at large } q \qquad (\text{for } |\mu| > 1), \tag{21}$$

a condition that is satisfied in most models of physical interest (including Models 1&2).

For general unitary invariant ensembles of the form $m_i = \Lambda_i V_i$, (V_i: Haar-random unitary; Λ_i: (positive) diagonal), we have $q = \left(\mathbb{E}\,\mathcal{P} - \frac{1}{q_{\text{tot}}} + \dots\right)^{-1}$, where $\mathcal{P} = \operatorname{tr}\Lambda^2/(\operatorname{tr}\Lambda)^2$ is the *purity* after a single timestep. (Caveat: $\mathbb{E}\,\mathcal{P}$ is dominated by rare events! [2])

Even for spatially local models without unitary invariance property, a coarse-graining argument shows that the scaling form holds.

References

[1] Collins, Benoit, Sho Matsumoto, and Jonathan Novak. "The weingarten calculus." NOTICES OF THE AMERICAN MATHEMATICAL SOCIETY (2022) 69, no. 5

[2] Fisher, Matthew PA, Vedika Khemani, Adam Nahum, and Sagar Vijay. "Random quantum circuits." Annual Review of Condensed Matter Physics 14, no. 1 (2023): 335-379.

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